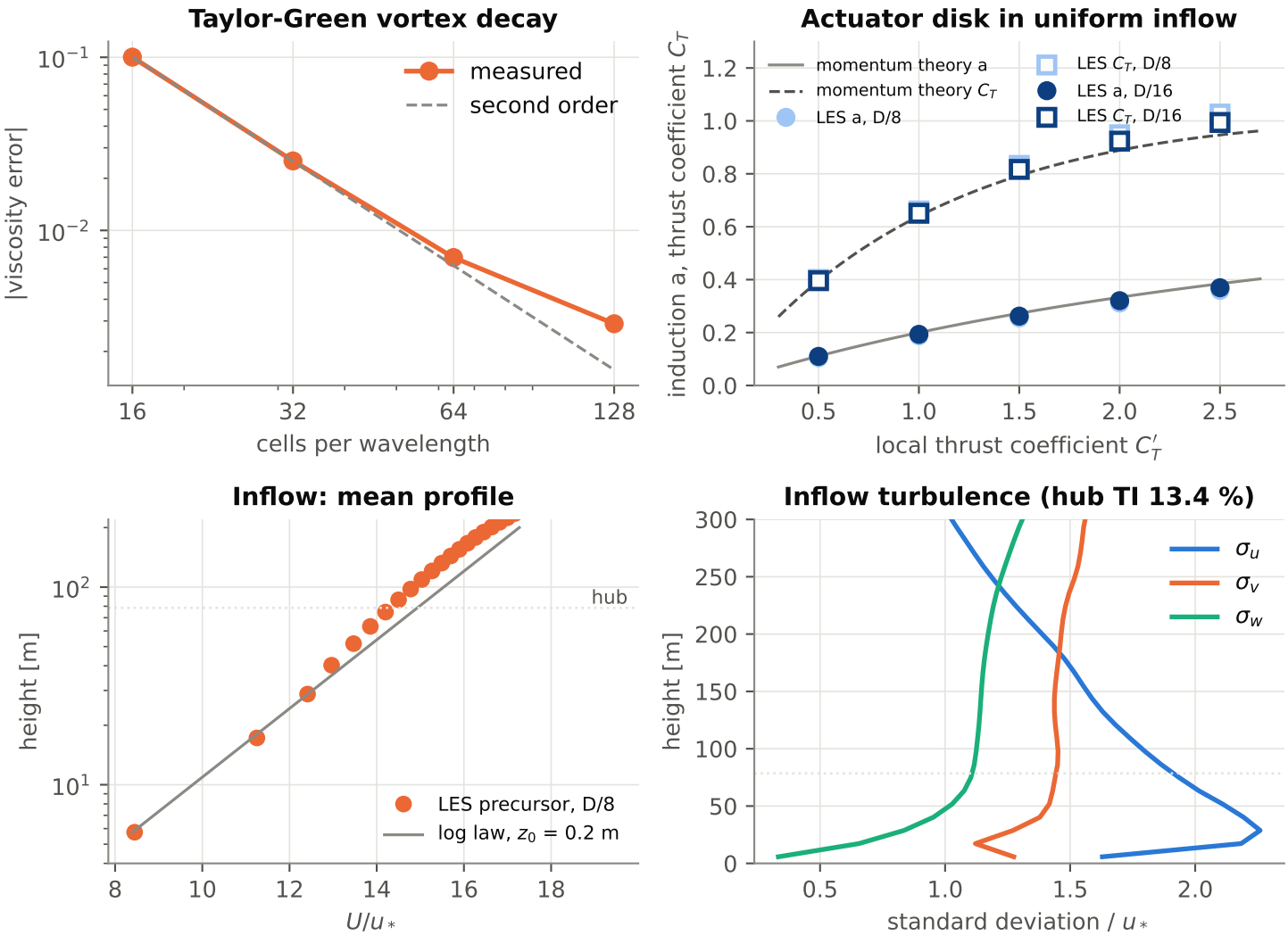
Taylor-Green vortex decay
measured
second order
|viscosity error|
10^{-1}
10^{-2}
16
32
64
128
cells per wavelength
Actuator disk in uniform inflow
momentum theory a
momentum theory C_T
LES a, D/8
LES C_T, D/8
LES a, D/16
LES C_T, D/16
induction a, thrust coefficient C_T
0.0
0.2
0.4
0.6
0.8
1.0
1.2
0.5
1.0
1.5
2.0
2.5
local thrust coefficient C_T'
Inflow: mean profile
hub
LES precursor, D/8
log law, z_0 = 0.2 m
height [m]
10^2
10^1
8
10
12
14
16
18
U/u_*
Inflow turbulence (hub TI 13.4 %)
σ_u
σ_v
σ_w
height [m]
0
50
100
150
200
250
300
0.5
1.0
1.5
2.0
standard deviation / u_*